
CAPSTONE PROJECT 2

Groundwater and Climate Data Analysis Project

Water Stress Across India using PostgreSQL, Python and Power BI

A complete end-to-end database management and analytics study of climate drivers, groundwater levels, seasonal recharge, depletion patterns and spatial groundwater risk across India.

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1. Introduction

India's groundwater system is shaped by climate variability, monsoon dependence, land use, extraction pressure and hydrogeological diversity. A purely rainfall-based view is not enough because the same rainfall can produce different groundwater outcomes depending on season, basin, state, aquifer type and previous groundwater level. This project therefore integrates location metadata, groundwater records and climate variables into a single relational and analytical workflow.

The project title, Water Stress Across India Using PostgreSQL, Python and Power BI, reflects three connected layers. PostgreSQL provides disciplined storage and relational integrity. Python provides reproducible cleaning, aggregation, statistical analysis and visualization. Power BI provides interactive business intelligence for comparing climate drivers and groundwater outcomes.

The final purpose is to identify where groundwater stress is most severe, how rainfall and temperature relate to groundwater behavior, which seasons drive recharge or depletion, and how decision-makers can use dashboards to monitor water stress.

Project Overview

This project analyzes the relationship between groundwater levels and climate variables across different locations in India. It combines groundwater observations from NITI Aayog with rainfall and temperature data from the India Meteorological Department (IMD), then uses SQL database design, Python EDA and Power BI dashboards to convert the data into technical findings and decision-ready insights.

Data Sources

- Groundwater Data: NITI Aayog.
- Climate Data (Rainfall and Temperature): India Meteorological Department (IMD).

2. Problem Statement and Objectives

Problem Statement

The project addresses the need for an integrated system that can store, analyze and communicate groundwater stress across Indian monitoring stations by combining location, climate and groundwater measurements. The challenge is not only to calculate averages, but also to reveal seasonal patterns, climate-groundwater relationships, outliers, geographic hotspots and dashboard-ready metrics.

Objectives

1. Design a normalized PostgreSQL database for groundwater and climate monitoring data.
2. Import the Excel dataset into PostgreSQL using pgAdmin-compatible table structures.
3. Clean, validate and aggregate station-month climate and groundwater records in Python.
4. Perform EDA covering distributions, categories, correlations, outliers, trends, seasonality and spatial risk.
5. Interpret every notebook visualization and connect results to groundwater sustainability.
6. Analyze both Power BI dashboards, including KPIs, slicers, filters, visuals, measures and business insights.
7. Document limitations, future scope and decision value for academic and operational use.

Data Collection

Groundwater data was collected from NITI Aayog and climate data, including rainfall and temperature variables, was collected from the India Meteorological Department (IMD). These datasets were integrated by station, year and month to support relational database storage, exploratory data analysis and dashboard reporting.

3. Dataset Description

The Excel workbook contains three sheets that directly represent the normalized database design. The groundwater observations are treated as NITI Aayog groundwater data, while rainfall and temperature variables are treated as IMD climate data. The location sheet contains station and geographic metadata, the groundwater sheet contains depth-to-water observations and groundwater status, and the climate sheet contains rainfall and temperature observations at station-month grain.

Dataset / Sheet	Rows	Purpose
Location_Master	17571	Station metadata, state, district, basin, coordinates and IMD matching fields
Groundwater_Data	125026	Groundwater readings, date, year, month, season, source, current level, previous level, change and status
Climate_Data	125026	Monthly rainfall, rainfall category, maximum/minimum temperature and temperature range
Master Analytical Dataset	107484	Station-year-month merged dataset used for EDA after aggregation

Key Coverage

- 17,571 unique monitoring stations.
- 30 states, 554 districts and 22 basins.
- Study period: 2019 to 2021.
- Zero missing cells and zero duplicate rows were detected in all three source sheets.

Column Structure

Sheet	Representative Columns	Column Count
Location_Master	station_id, state_code, state_name, district_code, district_name, station_name, basin, sub_basin ...	19
Groundwater_Data	gw_record_id, station_id, _id, date, year, month, season, source ...	12
Climate_Data	climate_record_id, station_id, year, month, rain_mm_monthly, rain_category, tmax_c_mean, tmin_c_mean ...	9

4. Database Design and ER Diagram

The ER diagram implements a one-to-many design. One station in Location_Master can have many groundwater records and many climate records. The station_id key acts as the bridge across tables.

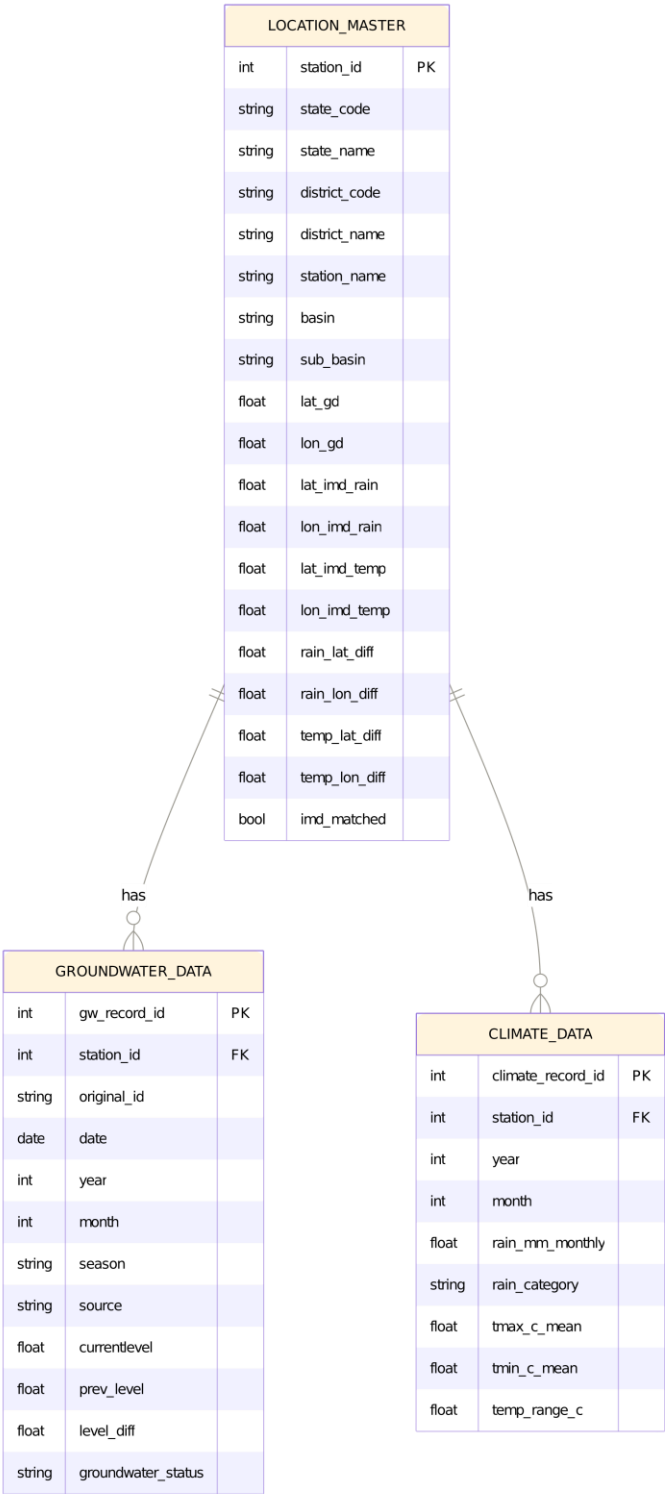


Figure 1: Entity Relationship Diagram showing Location_Master as the parent table and Groundwater_Data / Climate_Data as child fact tables.

Table	Primary Key	Foreign Key	Role
Location_Master	station_id	None	Master station, state, district, basin, coordinate and IMD matching metadata.
Groundwater_Data	gw_record_id	station_id references Location_Master(station_id)	Observed groundwater level, level change, season and status.
Climate_Data	climate_record_id	station_id references Location_Master(station_id)	Monthly rainfall category, rainfall amount and temperature variables.

Normalization Logic

- Station attributes are stored once in Location_Master instead of being repeated for every monthly observation.
- Groundwater and climate observations are separated because they represent different measurement processes and can evolve independently.
- Foreign key relationships preserve referential integrity: every observation must belong to a valid station.
- The design supports station-level joins, year-month analysis, state/district/basin aggregation and dashboard modeling.

5. PostgreSQL Implementation

The PostgreSQL implementation uses CREATE TABLE statements for the three normalized tables. Numeric fields are stored using DECIMAL precision where measurement values require stable formatting. Station and record identifiers are integer keys. The child tables include explicit foreign key constraints to Location_Master.

SQL Database Schema

The SQL database schema is included in Appendix A under the label SQL Query Placeholder. In this final report, the placeholder is replaced with the actual CREATE TABLE statements and validation queries from SQL_Queries.sql.

Implementation Workflow

8. Create a database in PostgreSQL using pgAdmin.
9. Execute SQL_Queries.sql to create Location_Master, Groundwater_Data and Climate_Data.
10. Import Excel sheets as CSV or use pgAdmin import/export with matching column order.
11. Load Location_Master first because child tables depend on station_id.
12. Load Groundwater_Data and Climate_Data after validating station_id values.
13. Run SELECT and COUNT validation queries to confirm table creation and import success.

6. Data Cleaning and Preprocessing

The notebook validates that all source tables are clean: no null cells and no duplicated source rows. The main preprocessing step is aggregation to station-year-month grain, because a small number of telemetric stations contain more frequent measurements. Aggregation avoids overweighting high-frequency stations while keeping climate and groundwater tables aligned.

Preprocessing Steps

14. Read all three Excel sheets using pandas.
15. Inspect shapes, data types, sample records and category values.
16. Validate missing values and duplicate rows.
17. Aggregate groundwater and climate observations by station_id, year and month.
18. Merge aggregated climate and groundwater data with location metadata.
19. Create a master analytical dataset with 107,484 station-month observations and 34 analytical fields.

Metric	Value	Interpretation
Highest recharge year	2019	Year with most negative average level change in the analytical dataset.
Average monthly rainfall	98.68 mm	Raw Climate_Data table KPI visible in Power BI.
Average maximum temperature	30.20 C	Raw Climate_Data table KPI visible in Power BI.
Extreme rainfall months	22.51%	Share of raw climate records classified as Extreme.
Average groundwater level	8.25 m	Raw Groundwater_Data table KPI visible in Power BI.
Average level change	-0.01 m	Raw Groundwater_Data table KPI visible in Power BI.

7. Exploratory Data Analysis

This chapter follows the Jupyter Notebook output sequence. Each figure below is extracted from the actual notebook output and is interpreted in terms of what it measures, how the visualization was produced, what pattern it reveals and why the pattern matters for water stress analysis.

Notebook Visual 1: Graph 1: Distribution of Depth-to-Water-Table

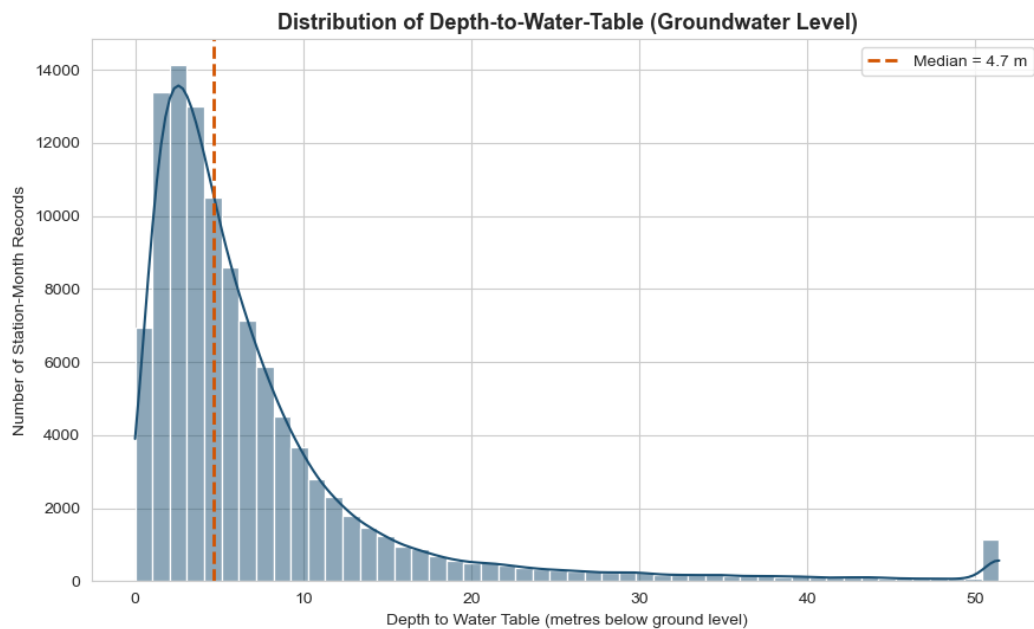


Figure 2: Graph 1: Distribution of Depth-to-Water-Table

Methodology

The notebook uses pandas to select the analytical variable and seaborn/matplotlib to plot a distribution. Extreme tails are clipped where needed to keep the main pattern readable.

Interpretation

- Distribution of Depth-to-Water-Table - What does this graph show?
- The frequency distribution of how deep the water table sits below ground level, across all 107,484 station-month readings.
- Key Findings: The distribution is heavily right-skewed.
- The median depth is 4.7m, and the bulk of readings fall between 1-10m.

Significance

The visual supports the project objective by translating raw monitoring records into interpretable patterns that can guide groundwater risk monitoring.

Notebook Visual 2: Graph 2: Groundwater Status Classification

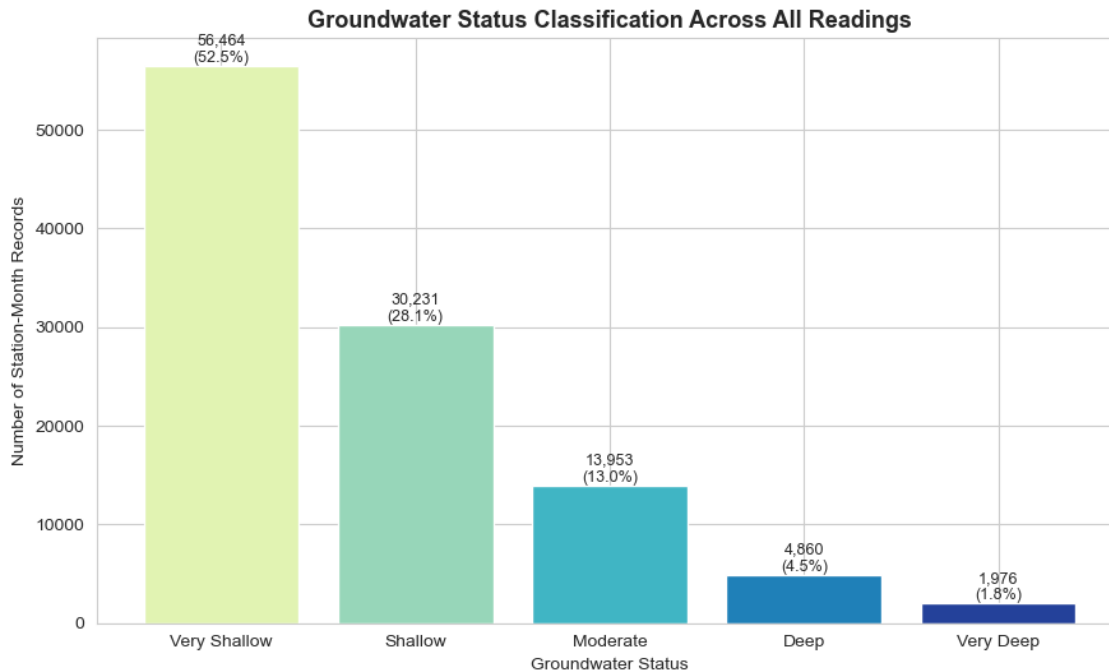


Figure 3: Graph 2: Groundwater Status Classification

Methodology

The notebook groups records by category, counts station-month observations and displays the result as a categorical bar or share chart.

Interpretation

- Groundwater Status Classification - What does this graph show?
- The count of station-month records in each CGWB-style severity bucket: Very Shallow to Very Deep.
- Key Findings: 80.8% of readings fall in "Very Shallow" or "Shallow" (healthy) categories.
- However, 1.8% ("Very Deep") and 4.5% ("Deep") represent genuinely stressed aquifers - a small percentage that still equates to hundreds of stations and the communities they serve.

Significance

The visual supports the project objective by translating raw monitoring records into interpretable patterns that can guide groundwater risk monitoring.

Notebook Visual 3: Graph 3: Rainfall Category Share

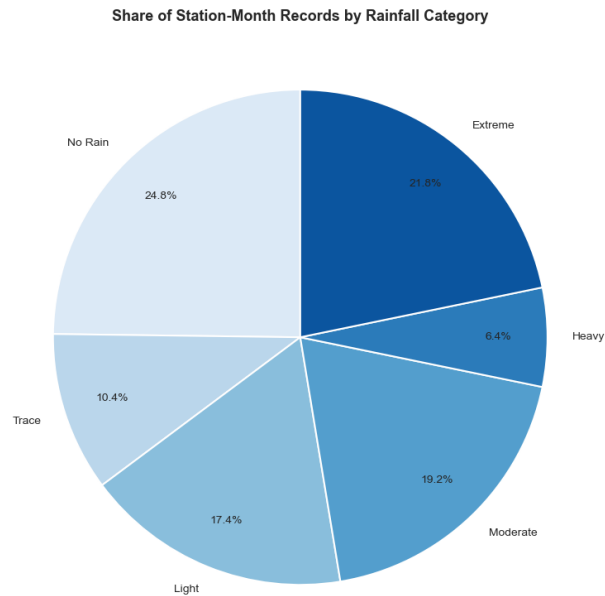


Figure 4: Graph 3: Rainfall Category Share

Methodology

The notebook groups records by category, counts station-month observations and displays the result as a categorical bar or share chart.

Interpretation

- Rainfall Category Share - What does this graph show?
- The proportion of station-months falling into each rainfall intensity bucket.
- Key Findings: "No Rain" is the single largest category (24.8%), closely followed by "Extreme" (21.8%) - a bimodal pattern rather than a smooth gradient.
- Possible Reasons: India's monsoon-dominated climate produces exactly this bimodal signature: long dry seasons (Oct-May) generate "No Rain"/"Trace" readings, while the concentrated monsoon (Jun-Sep) generates "Heavy"/"Extreme" readings - there is comparatively little "Moderate, steady rainfall" in between.

Significance

Rainfall is a primary recharge driver, but the charts show that rainfall alone cannot explain groundwater stress. Its timing, intensity and regional distribution must be interpreted with depth-to-water and seasonal change.

Notebook Visual 4: Graph 4: Box Plot - Monthly Rainfall

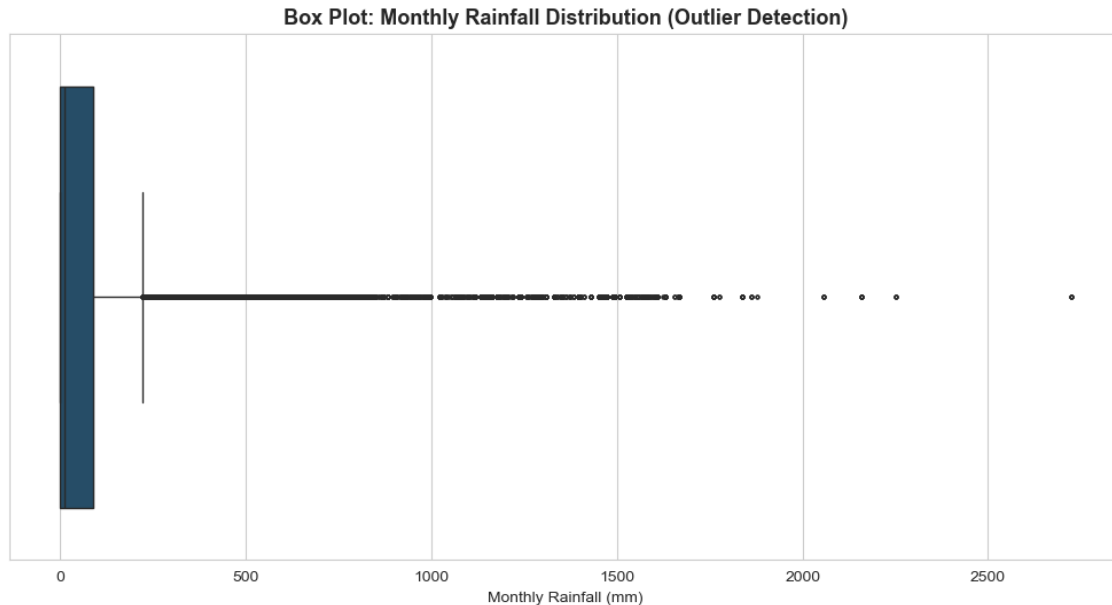


Figure 5: Graph 4: Box Plot - Monthly Rainfall

Methodology

The notebook uses a box plot to show median, interquartile range and outliers. This is appropriate for identifying skew and extreme measurement events.

Interpretation

- Box Plot - Monthly Rainfall - What does this graph show?
- The spread, median, and outliers of monthly rainfall across all station-months.
- Key Findings: The median sits near 0-20mm with the interquartile range compact, but a long right tail of outlier points extends past 1,500mm/month.
- Possible Reasons: Reflects the same dry-season/monsoon bimodality as Graph 3 - most months are dry-season, so the "typical" box is near zero, while monsoon-month outliers represent genuine extreme rainfall events (not data errors, confirmed in Section 7).

Significance

Rainfall is a primary recharge driver, but the charts show that rainfall alone cannot explain groundwater stress. Its timing, intensity and regional distribution must be interpreted with depth-to-water and seasonal change.

Notebook Visual 5: Graph 5: Density Plot - Max Temperature

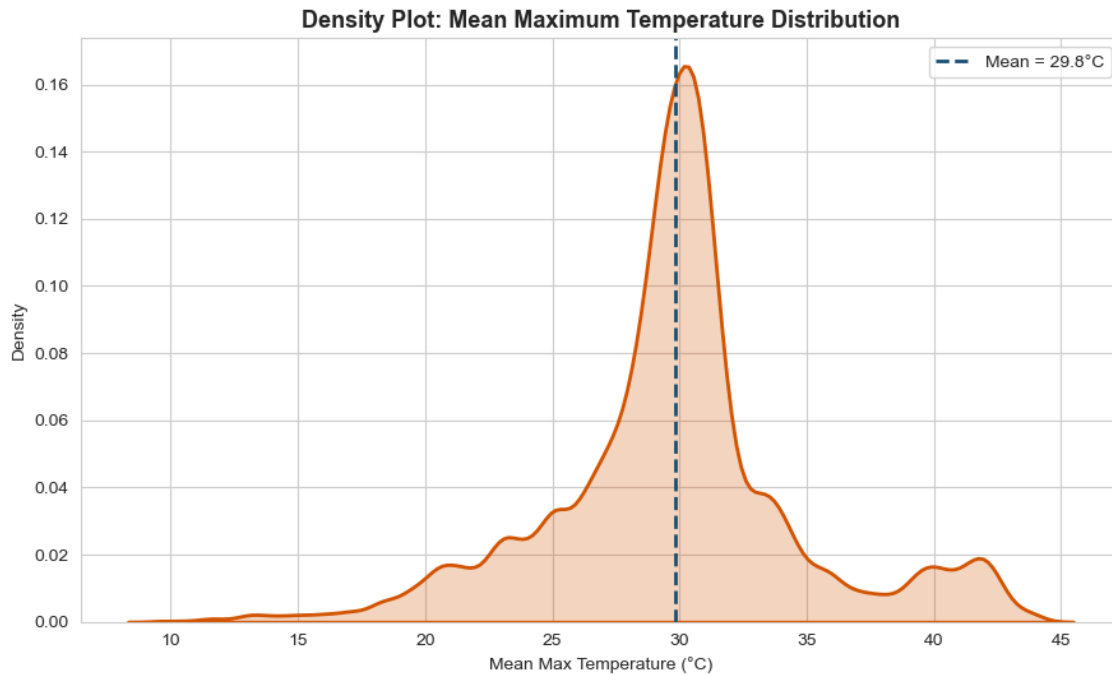


Figure 6: Graph 5: Density Plot - Max Temperature

Methodology

The notebook uses a kernel density estimate to show the smoothed distribution of monthly maximum temperature.

Interpretation

- Density Plot - Max Temperature - What does this graph show?
- The smoothed probability density of mean monthly maximum temperatures.
- Key Findings: The distribution is roughly bimodal/broad, averaging 29.8 deg C, with a secondary bump in the high-30s deg C range corresponding to peak summer (April-June, pre-monsoon).
- Possible Reasons: India's strong seasonal temperature cycle - cool winters, scorching pre-monsoon summers - produces this dual-peak structure rather than a single bell curve.

Significance

Temperature is important because higher heat can increase evapotranspiration and irrigation demand, indirectly increasing pressure on groundwater even when rainfall occurs.

Notebook Visual 6: Graph 6: Distribution of Water Table Change

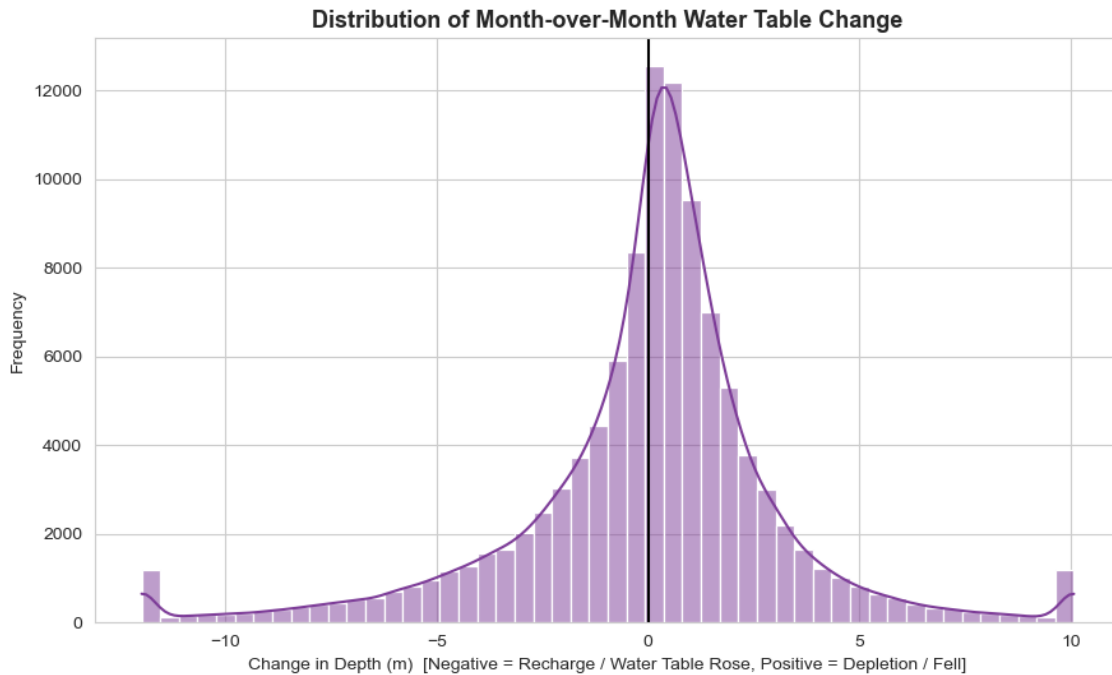


Figure 7: Graph 6: Distribution of Water Table Change

Methodology

The notebook uses pandas to select the analytical variable and seaborn/matplotlib to plot a distribution. Extreme tails are clipped where needed to keep the main pattern readable.

Interpretation

- Distribution of Water Table Change - What does this graph show?
- How much the depth-to-water changes between consecutive readings at the same station.
- Key Findings: The distribution centers near zero but is slightly right-skewed (more mass on the depletion/positive side), suggesting depletion events are both common and occasionally severe.
- Possible Reasons: Net long-term extraction exceeding recharge in a meaningful subset of stations, consistent with national over-exploitation concerns documented by CGWB.

Significance

Level change is one of the most decision-relevant indicators because it separates recharge periods from depletion periods and helps identify seasonal stress windows.

Notebook Visual 7: Graph 7: Monitoring Coverage by State

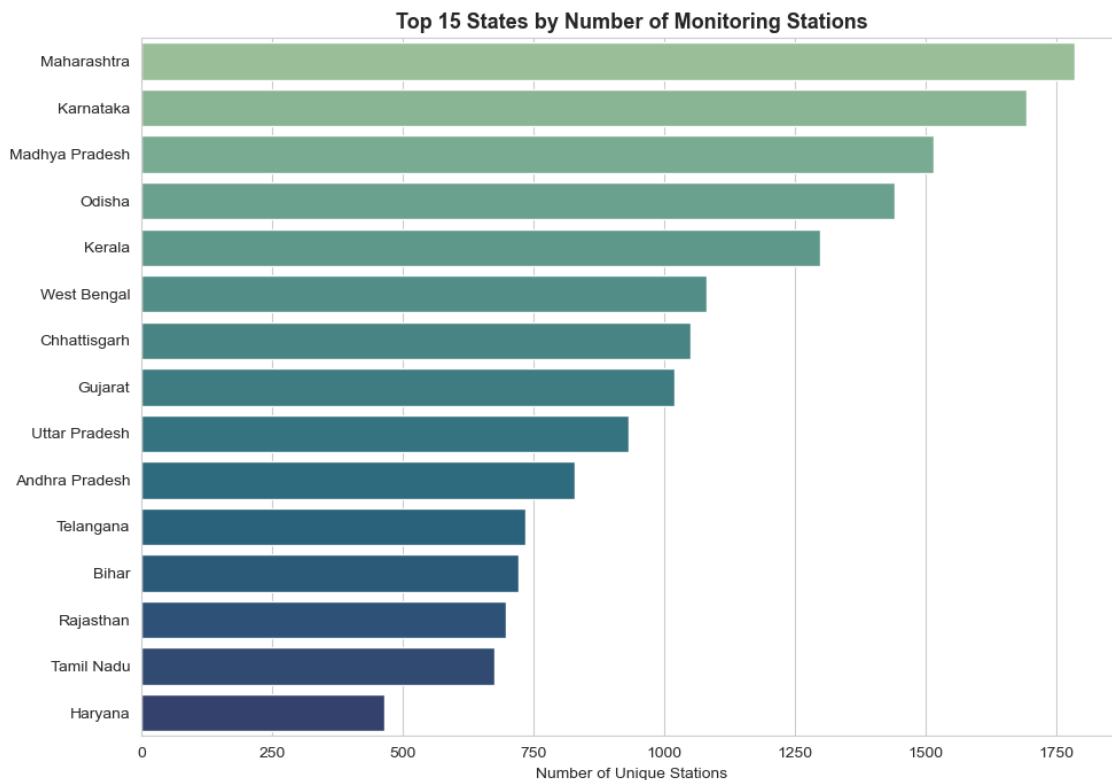


Figure 8: Graph 7: Monitoring Coverage by State

Methodology

The notebook applies pandas aggregation and seaborn/matplotlib visualization to convert station-month records into an interpretable analytical view.

Interpretation

- Monitoring Coverage by State - What does this graph show?
- Which states have the densest groundwater monitoring networks.
- Key Findings: Karnataka, Madhya Pradesh, and Gujarat have the highest station counts, each with 1,000+ monitoring points.
- Possible Reasons: Reflects both genuine state size/agricultural intensity and historical investment in monitoring infrastructure by state groundwater boards.

Significance

Spatial analysis converts large monitoring data into actionable hotspots. State, district and basin views help prioritize where water management attention is most needed.

Notebook Visual 8: Graph 8: Temperature Distribution by Season

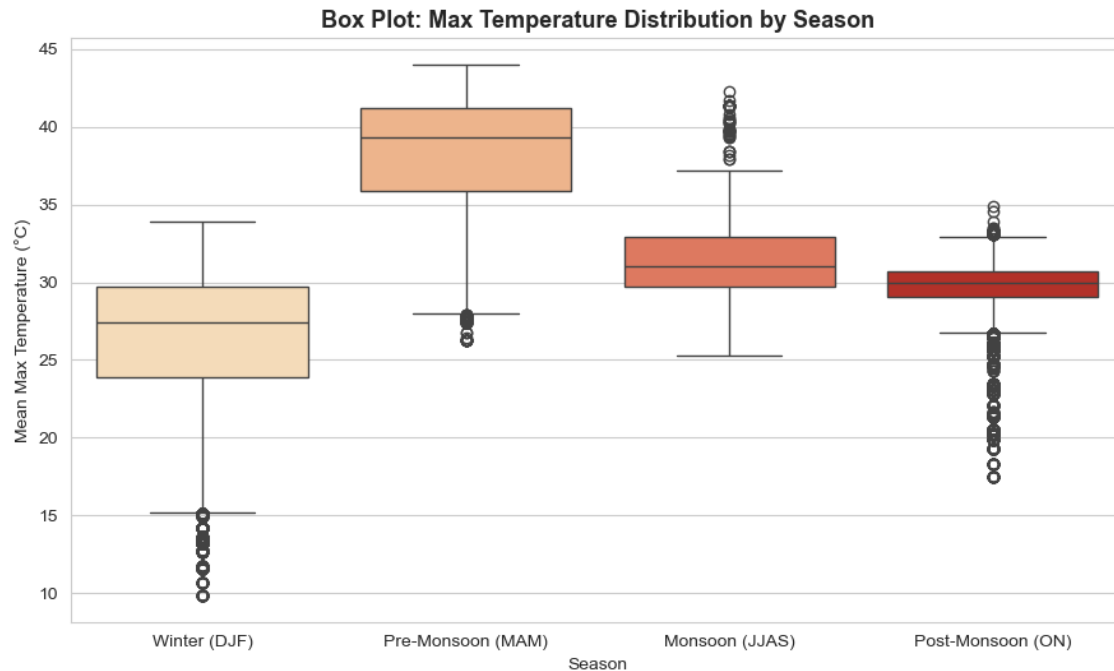


Figure 9: Graph 8: Temperature Distribution by Season

Methodology

The notebook uses pandas to select the analytical variable and seaborn/matplotlib to plot a distribution. Extreme tails are clipped where needed to keep the main pattern readable.

Interpretation

- Temperature Distribution by Season - What does this graph show?
- How maximum temperature varies and spreads across the four meteorological seasons.
- Key Findings: Pre-Monsoon (MAM) is both the hottest (~38 deg C median) and most variable season; Winter is coolest and most consistent.
- Possible Reasons: Standard pre-monsoon heat-buildup pattern across the Indian subcontinent before monsoon onset cools the surface.

Significance

Temperature is important because higher heat can increase evapotranspiration and irrigation demand, indirectly increasing pressure on groundwater even when rainfall occurs.

Notebook Visual 9: Graph 9: Depth-to-Water by State & Season

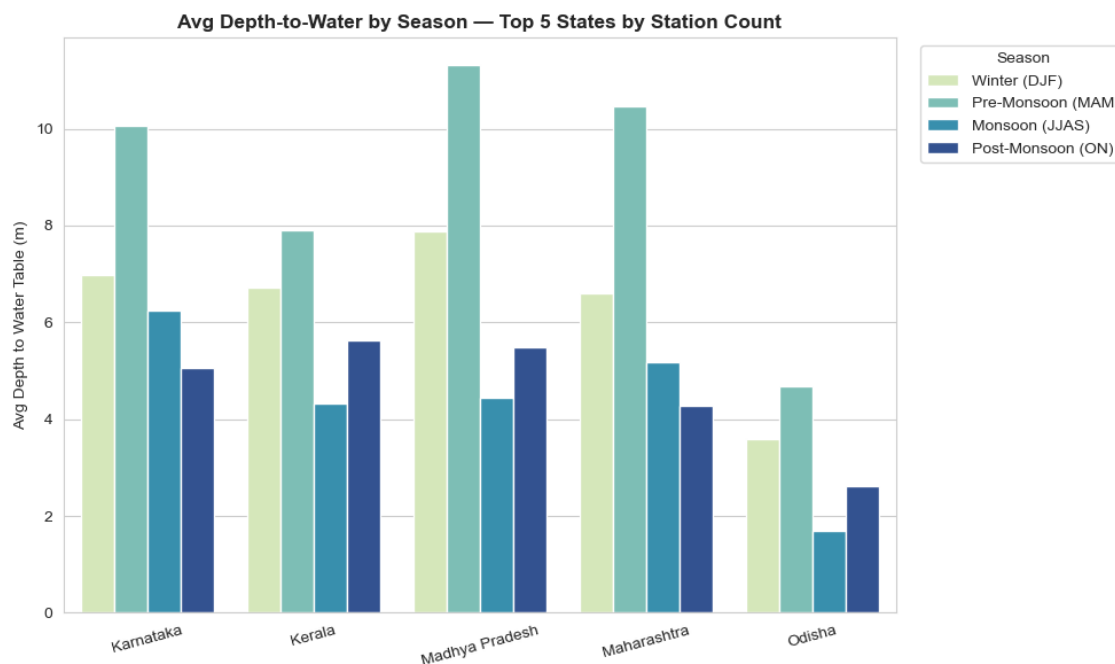


Figure 10: Graph 9: Depth-to-Water by State & Season

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- Depth-to-Water by State & Season - Relationship Identified: Seasonal swing magnitude varies substantially by state, not just by season.
- Positive or Negative: Karnataka shows the largest seasonal range (deepest in Pre-Monsoon, shallowest in Monsoon); Madhya Pradesh shows a more muted swing.
- Trends: Every one of the top-5 states follows the same Monsoon-shallow / Pre-Monsoon-deep pattern, but the amplitude differs - a signal of differing aquifer storage capacity and extraction intensity.
- Potential Causes: Hard-rock aquifers (common in Karnataka's Deccan terrain) have lower storage capacity and react more sharply to seasonal rainfall than the alluvial aquifers common in parts of Madhya Pradesh.

Significance

Spatial analysis converts large monitoring data into actionable hotspots. State, district and basin views help prioritize where water management attention is most needed.

Notebook Visual 10: Graph 10: Water Table Change by Season

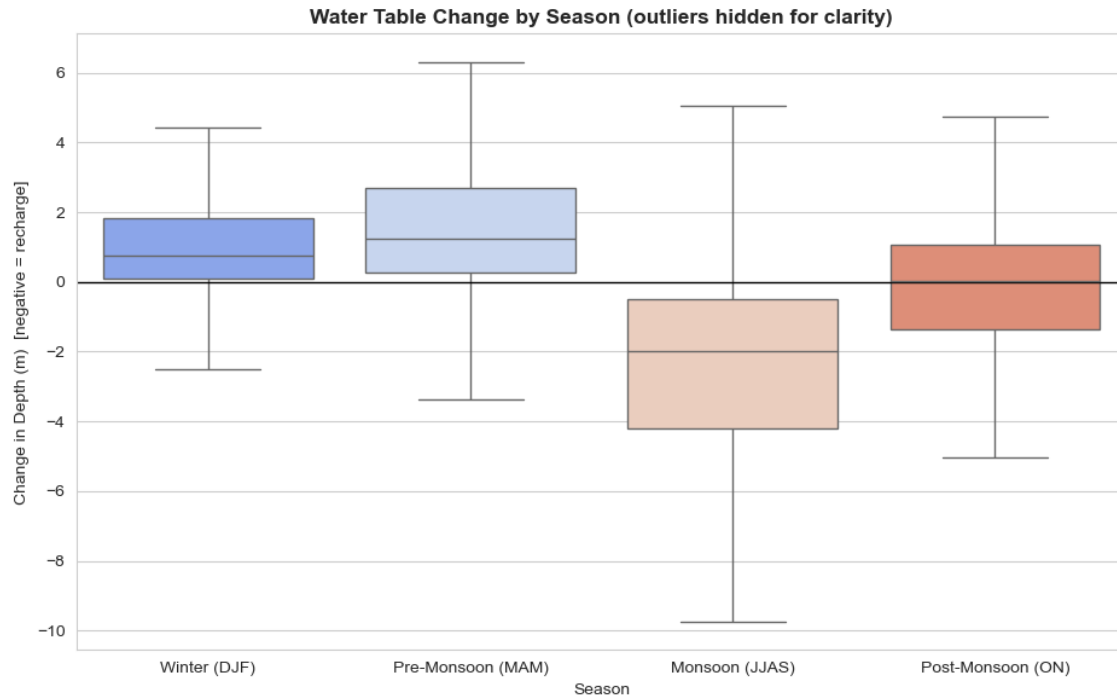


Figure 11: Graph 10: Water Table Change by Season

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- Water Table Change by Season - Relationship Identified: A clean, strong seasonal signal - Monsoon medians sit decisively below zero (recharge); Winter and Pre-Monsoon sit above zero (depletion).
- Positive or Negative: Monsoon: negative (recharge).
- Pre-Monsoon: most positive (worst depletion).
- Trends: The boxes barely overlap between Monsoon and Pre-Monsoon - this is one of the strongest, most consistent patterns in the entire dataset.

Significance

Level change is one of the most decision-relevant indicators because it separates recharge periods from depletion periods and helps identify seasonal stress windows.

Notebook Visual 11: Graph 11: Rainfall Received by Groundwater Status

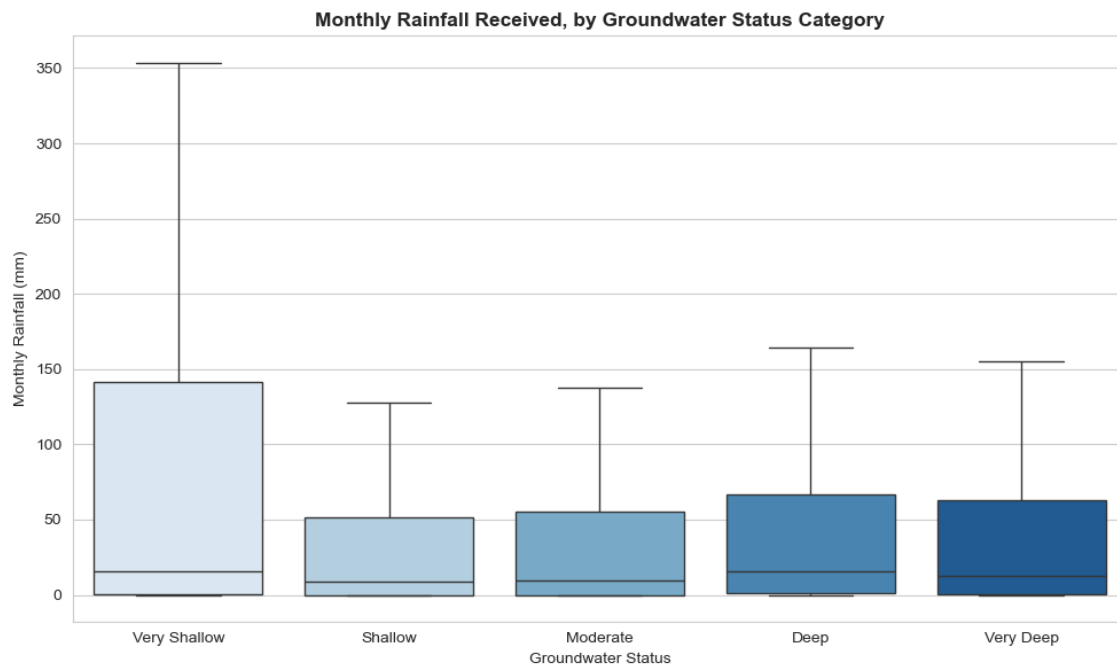


Figure 12: Graph 11: Rainfall Received by Groundwater Status

Methodology

The notebook applies pandas aggregation and seaborn/matplotlib visualization to convert station-month records into an interpretable analytical view.

Interpretation

- Rainfall Received by Groundwater Status - Relationship Identified: Rainfall received is broadly similar across all groundwater status categories, with only modest differences in median.
- Positive or Negative: Slight negative drift - "Very Deep" stations show marginally lower typical rainfall than "Very Shallow" stations, but with substantial overlap.
- Trends: This is a cross-sectional snapshot, not a time trend - it compares different stations' current rainfall against their current status, which reflects years of accumulated extraction history, not just this period's weather.
- Potential Causes: Groundwater status is a function of cumulative long-term extraction vs.

Significance

Rainfall is a primary recharge driver, but the charts show that rainfall alone cannot explain groundwater stress. Its timing, intensity and regional distribution must be interpreted with depth-to-water and seasonal change.

Notebook Visual 12: Graph 12: Correlation Matrix

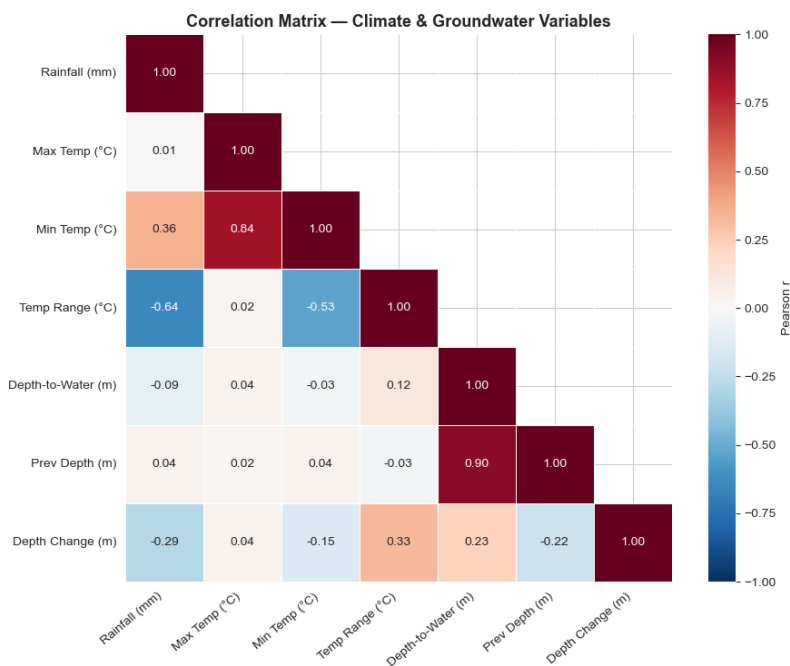


Figure 13: Graph 12: Correlation Matrix

Methodology

The notebook computes Pearson correlations across numerical climate and groundwater variables and visualizes the correlation matrix as a heat map.

Interpretation

- Correlation Matrix - Strongest Relationships: Prev Depth and Depth-to-Water ($r=0.90$, expected - depth is highly autocorrelated month-to-month).
- Among *distinct* drivers: Temp Range and Depth Change ($r=0.33$) and Rainfall and Depth Change ($r=-0.29$) are the two strongest external drivers of groundwater change.
- Weakest Relationships: Max Temp and Rainfall ($r=0.01$) and Max Temp and Depth-to-Water ($r=0.04$) - confirming Graph 12's finding that temperature is a weak *direct* driver nationally.
- Hidden Patterns: Temp Range (the gap between daily max and min) is more strongly linked to depth change ($r=0.33$) than either temperature extreme alone - wide temp swings (typical of dry-season clear skies) associate with depletion, while narrow swings (typical of cloudy monsoon conditions) associate with recharge.

Significance

Multivariate analysis shows that groundwater behavior is not controlled by one factor. Lag, season, prior depth and regional conditions must be considered together.

Notebook Visual 13: Graph 13: Pairplot by Season

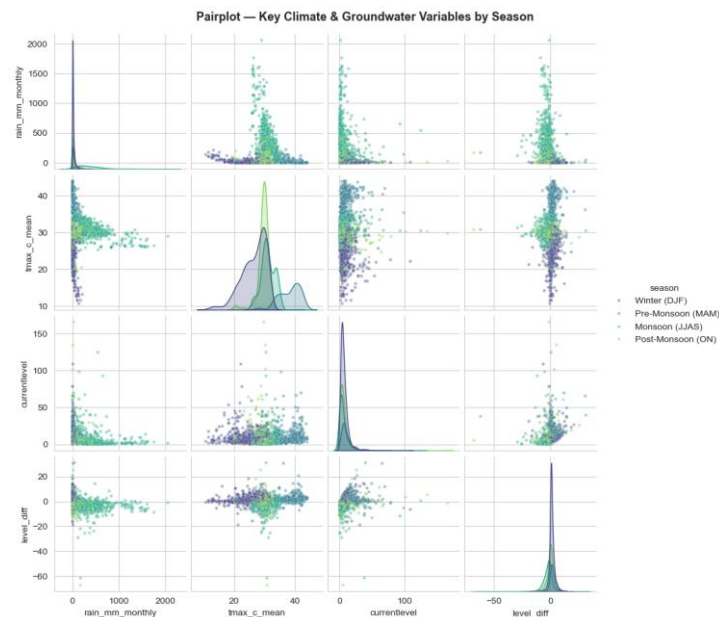


Figure 14: Graph 13: Pairplot by Season

Methodology

The notebook samples observations and plots pairwise relationships by season, allowing multivariate patterns and seasonal clusters to be inspected together.

Interpretation

- **Pairplot by Season - Strongest Relationships:** The `rain_mm_monthly` x `level_diff` panel shows the clearest season-separated clustering - Monsoon (typically shown in one consistent hue) clusters distinctly toward high-rain/negative-change, separated from the other three seasons.
- **Weakest Relationships:** `tmax_c_mean` panels show heavy overlap across all seasons except Pre-Monsoon, which separates out toward the high-temperature end - consistent with Graph 8.
- **Hidden Patterns:** The diagonal KDE curves reveal that `level_diff` is the only variable with meaningfully different *shape* (not just shifted location) across seasons - Monsoon's distribution is wider, reflecting more heterogeneous recharge outcomes station-to-station.
- **Unexpected Findings:** Even within the Monsoon season cluster, a meaningful subset of points still show positive `level_diff` (depletion) despite high rainfall - these are candidate "monsoon failure to recharge" stations worth individual investigation (likely chronically over-pumped or geologically poor-storage sites).

Significance

Multivariate analysis shows that groundwater behavior is not controlled by one factor. Lag, season, prior depth and regional conditions must be considered together.

Notebook Visual 14: Graph 14: Yearly Trend - Rainfall vs. Depth

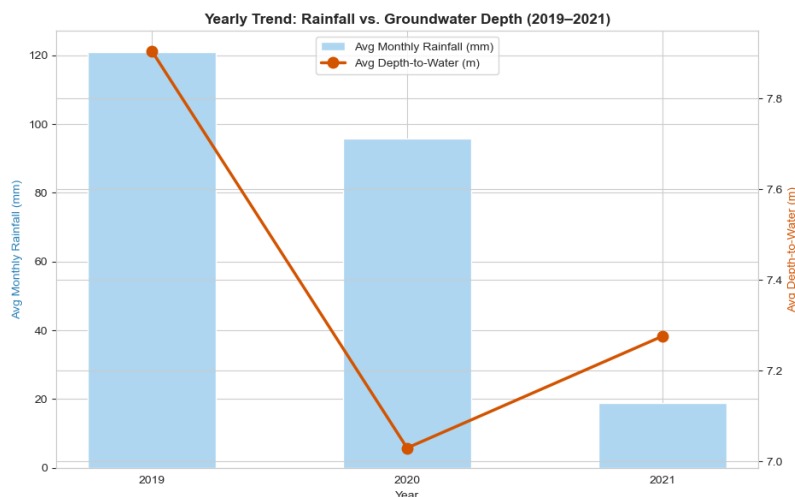


Figure 15: Graph 14: Yearly Trend - Rainfall vs. Depth

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- **Depth - Trend Direction:** Average recorded rainfall declined sharply across the three years (121mm to 96mm to 19mm monthly average), while average depth-to-water improved from 2019 (7.9m) to 2020 (7.0m) before edging back up slightly in 2021 (7.3m).
- **Growth/Decline Patterns:** The rainfall decline is partly a station-coverage artifact - fewer/different stations reporting in 2021 will shift the average; this should be read directionally, not as a precise national rainfall trend (note the 2021 figure is also affected by partial-year reporting in the underlying data).
- **Possible Causes:** Real year-to-year monsoon variability is the most likely primary driver; reporting-coverage changes are a secondary contributor that should be validated against IMD's official national rainfall figures before being used in external communications.
- **Key Observation:** Depth-to-water did not worsen in lockstep with the apparent rainfall decline - suggesting extraction behavior (e.g., reduced agricultural activity during 2020 COVID disruptions) may have partially offset reduced recharge.

Significance

Rainfall is a primary recharge driver, but the charts show that rainfall alone cannot explain groundwater stress. Its timing, intensity and regional distribution must be interpreted with depth-to-water and seasonal change.

Notebook Visual 15: Graph 15: Monthly Seasonality

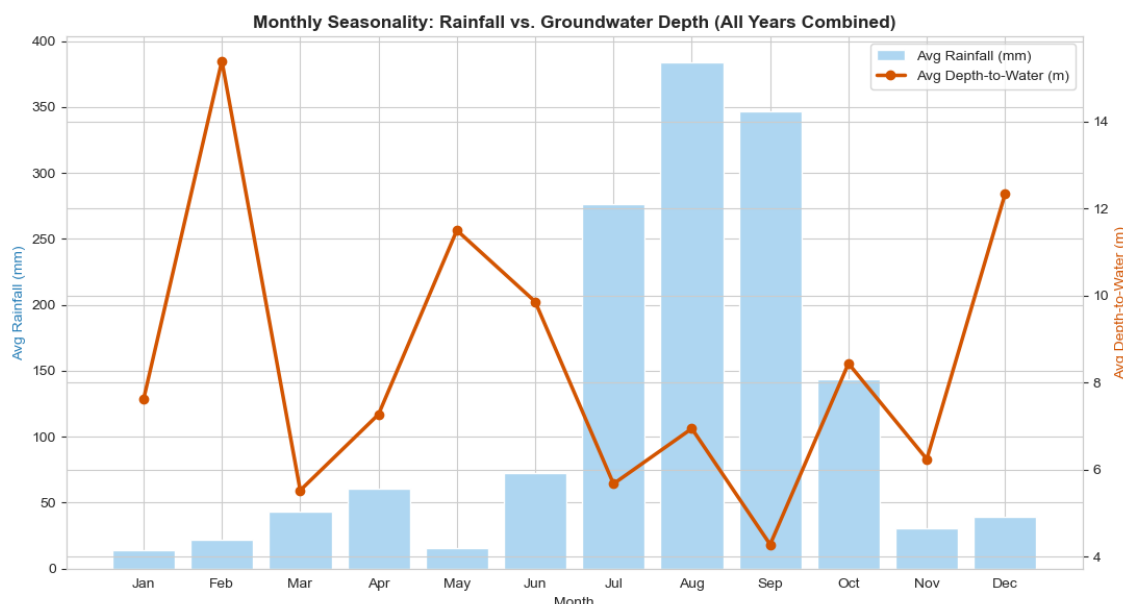


Figure 16: Graph 15: Monthly Seasonality

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- **Monthly Seasonality - Trend Direction:** A clear lagged-response pattern - rainfall peaks Jul-Sep, while the water table reaches its shallowest (best) point slightly after, and its deepest (worst) point in the pre-monsoon months (Feb, May).
- **Growth/Decline Patterns:** Note the data is station-readings-weighted (quarterly CGWB readings cluster around Jan/Aug/Nov per the typical 3-readings-per-year schedule), so apparent month-to-month "noise" partly reflects which stations report in which month, not pure signal.
- **Possible Causes:** Physical recharge lag - rain must percolate through soil before reaching the water table, typically a multi-week to multi-month delay depending on soil/rock type.
- **Key Observation:** The February reading (post-Winter, pre-Pre-Monsoon) shows an anomalous shallow reading worth noting in the data - likely reflects a specific CGWB measurement-cycle timing rather than a real recharge event.

Significance

The visual supports the project objective by translating raw monitoring records into interpretable patterns that can guide groundwater risk monitoring.

Notebook Visual 16: Graph 16: Seasonal Recharge vs. Depletion

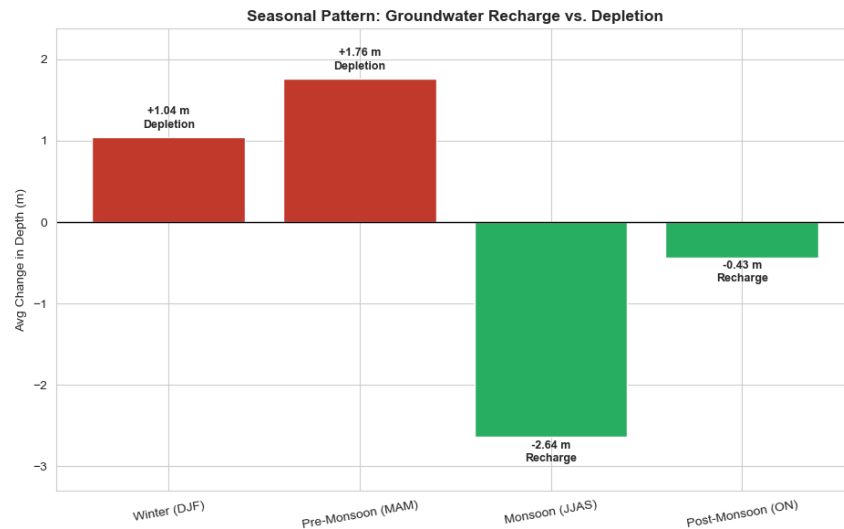


Figure 17: Graph 16: Seasonal Recharge vs. Depletion

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- **Depletion - Trend Direction:** Monsoon delivers strong net recharge (−2.64m average rise); Pre-Monsoon shows the steepest depletion (+1.76m average fall); Winter and Post-Monsoon are moderate depletion/recharge respectively.
- **Growth/Decline Patterns:** The Monsoon recharge (−2.64m) is roughly 1.5x larger in magnitude than the worst depletion season (+1.76m) - meaning a "normal" monsoon should be more than sufficient to offset typical dry-season drawdown, IF extraction stays within historical norms.
- **Possible Causes:** This is the clearest, most textbook seasonal hydrological pattern in the dataset, directly validating the Section 5/6 rainfall-recharge relationship at full strength when isolated by season rather than diluted across all months together.
- **Key Observation:** Any year where Monsoon recharge falls meaningfully short of −2.64m (a below-normal monsoon) is associated with elevated Pre-Monsoon-season stress in the historical record.

Significance

Level change is one of the most decision-relevant indicators because it separates recharge periods from depletion periods and helps identify seasonal stress windows.

Notebook Visual 17: Graph 17: Quarterly Time-Series

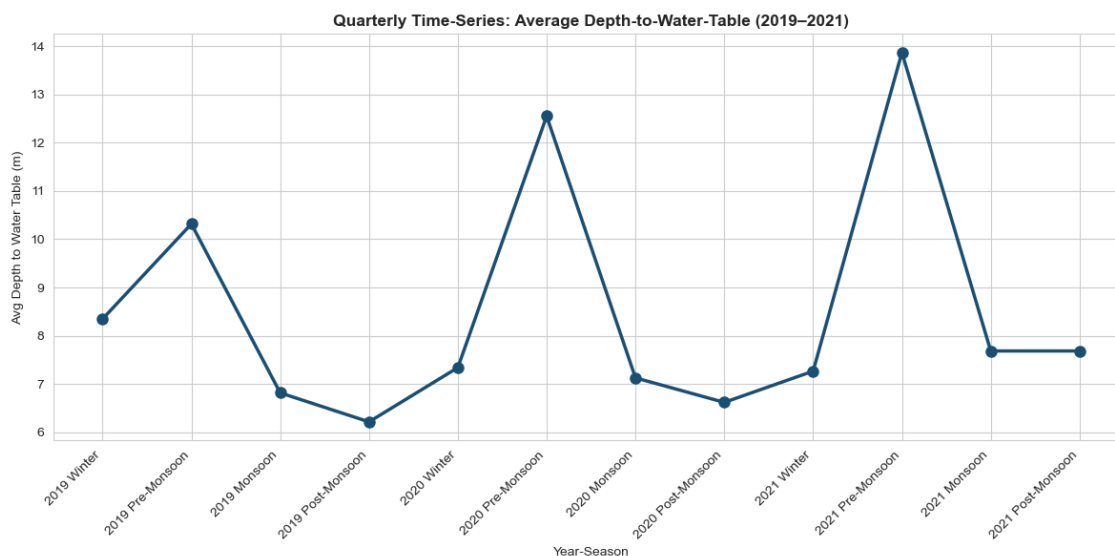


Figure 18: Graph 17: Quarterly Time-Series

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- **Quarterly Time-Series - Trend Direction:** A consistent zig-zag (sawtooth) pattern repeats across all three years - depth rises into Pre-Monsoon, drops sharply at Monsoon, partially recovers, then rises again - the seasonal cycle dominates any longer-term multi-year trend.
- **Growth/Decline Patterns:** No clear multi-year worsening or improving trend is visible in this 3-year window at the national aggregate level - the seasonal cycle amplitude is far larger than any year-over-year drift, consistent with Graph 16.
- **Possible Causes:** A national average necessarily smooths over the state-level depletion trends already identified in Section 9 (e.g., Haryana, Punjab) - a flat national line can coexist with serious sub-national depletion trends.
- **Key Observation:** Three years is too short a window to confidently distinguish a genuine long-term depletion trend from normal monsoon-to-monsoon variability at the national level - but it is not too short at the state/district level where multi-year CGWB records already exist.

Significance

The visual supports the project objective by translating raw monitoring records into interpretable patterns that can guide groundwater risk monitoring.

Notebook Visual 18: Map 18: Groundwater Stress Risk Map

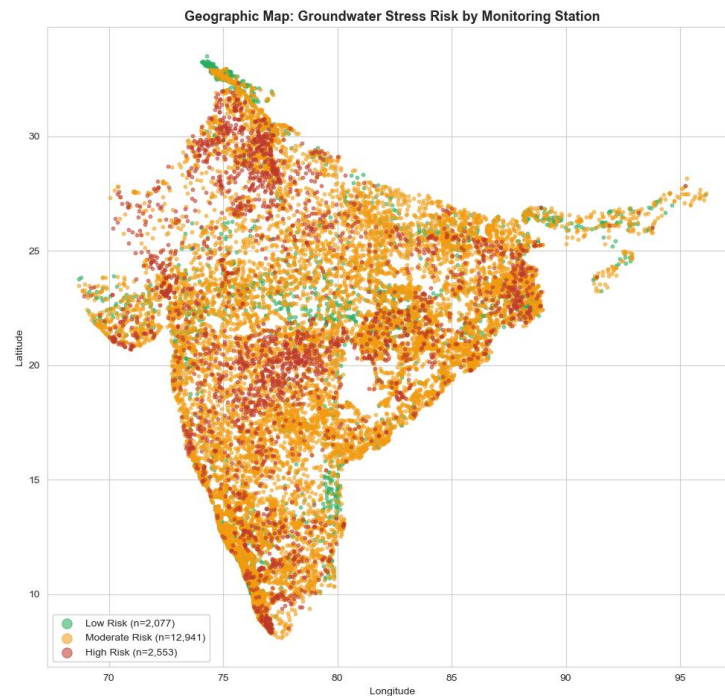


Figure 19: Map 18: Groundwater Stress Risk Map

Methodology

The notebook aggregates records to station level and plots each station using latitude and longitude coordinates, coloring points by the selected risk or climate variable.

Interpretation

- Map 18: Groundwater Stress Risk Map - What the map shows: Every monitoring station plotted at its true latitude/longitude, colored by risk tier (green=low, amber=moderate, red=high).
- High-performing (low-risk) regions: Clear green clustering in the northeast (Assam, Meghalaya), Jammu & Kashmir, and parts of the Himalayan foothills.
- Low-performing (high-risk) regions: Visible red clustering in the northwest corridor (Punjab-Haryana-western UP) and scattered through the Deccan plateau interior (Telangana, parts of Maharashtra/Karnataka).
- Geographic clusters: The northwest red cluster is the most visually dense and contiguous high-risk zone in the country - consistent with India's well-documented Green-Revolution-era groundwater over-exploitation in Punjab/Haryana.

Significance

Spatial analysis converts large monitoring data into actionable hotspots. State, district and basin views help prioritize where water management attention is most needed.

Notebook Visual 19: Map 19: Depth-to-Water Heat Map

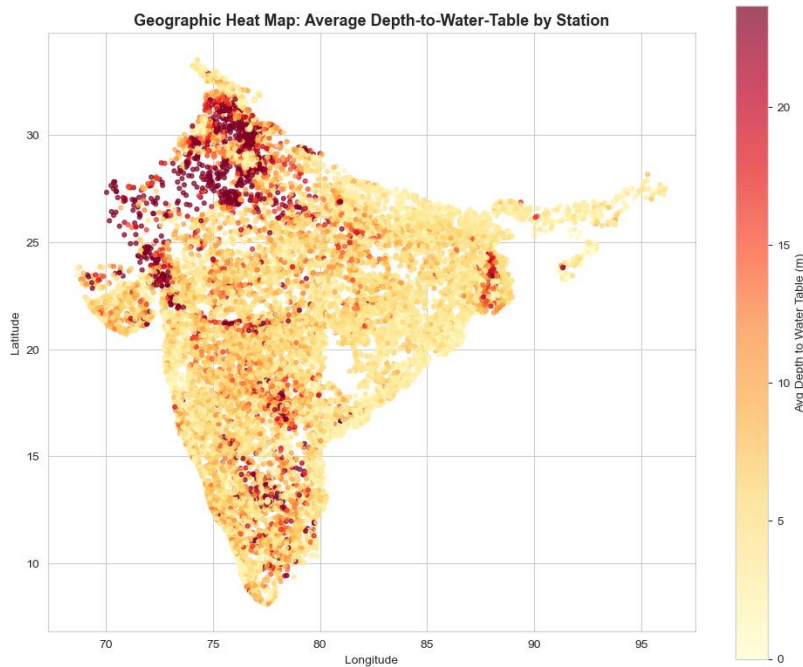


Figure 20: Map 19: Depth-to-Water Heat Map

Methodology

The notebook aggregates records to station level and plots each station using latitude and longitude coordinates, coloring points by the selected risk or climate variable.

Interpretation

- Map 19: Depth-to-Water Heat Map - What the map shows: A continuous-color view of average water table depth - darker red indicates deeper (more stressed) water tables.
- High-performing regions: The Indo-Gangetic plain (UP, Bihar, West Bengal) shows predominantly pale/light shading, reflecting naturally shallow alluvial aquifers.
- Low-performing regions: Pockets of dark red appear across central-western India, particularly Gujarat and parts of Rajasthan, where some stations average 60m+ depth.
- Geographic clusters: Unlike the risk map (which blends multiple factors), this single-variable view confirms depth itself is geologically clustered - alluvial-plain states are structurally advantaged regardless of management practices.

Significance

Spatial analysis converts large monitoring data into actionable hotspots. State, district and basin views help prioritize where water management attention is most needed.

Notebook Visual 20: Map 19: Rainfall Distribution Map

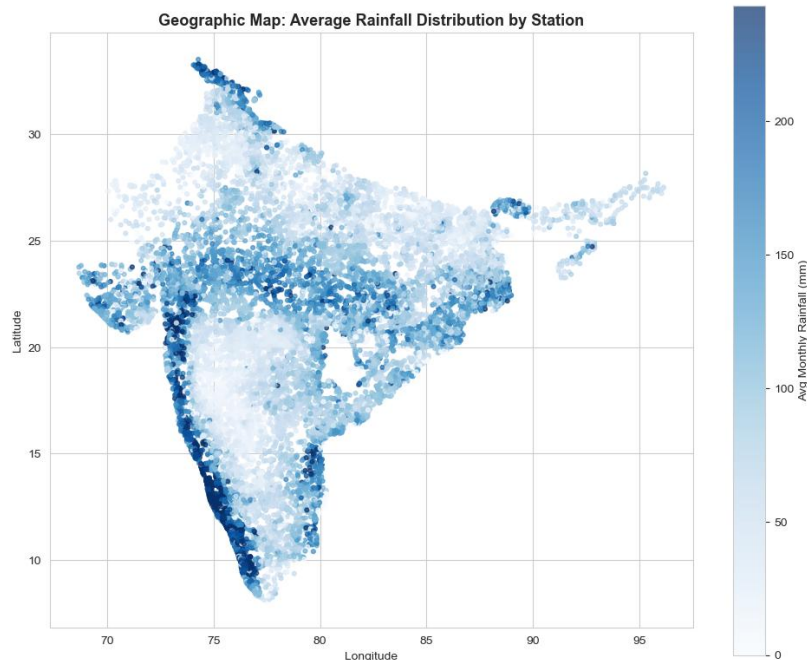


Figure 21: Map 19: Rainfall Distribution Map

Methodology

The notebook uses pandas to select the analytical variable and seaborn/matplotlib to plot a distribution. Extreme tails are clipped where needed to keep the main pattern readable.

Interpretation

- Map 19: Rainfall Distribution Map - What the map shows: Average monthly rainfall received by each station's location, dark blue = highest rainfall.
- High-performing regions: The Western Ghats (west coast) and Northeast India show the darkest blue - among the highest rainfall zones on Earth.
- Low-performing regions: The Deccan plateau interior (rain-shadow east of the Western Ghats) shows pale/near-white shading - a genuine climatological dry zone.
- Geographic clusters: This map closely matches well-documented Indian Meteorological Department rainfall climatology, serving as a strong validation check that the dataset's geography and climate data are internally consistent and realistic.

Significance

Rainfall is a primary recharge driver, but the charts show that rainfall alone cannot explain groundwater stress. Its timing, intensity and regional distribution must be interpreted with depth-to-water and seasonal change.

Notebook Visual 21: Graph 20: State-Level Risk Ranking

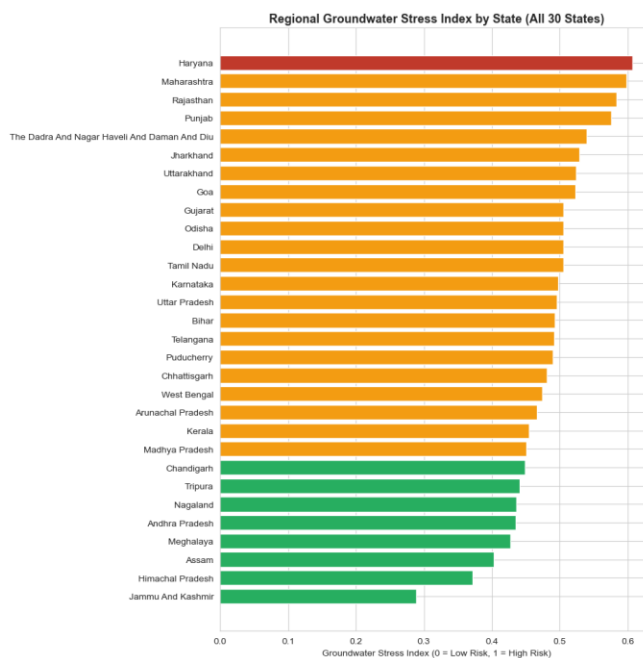


Figure 22: Graph 20: State-Level Risk Ranking

Methodology

The notebook applies pandas aggregation and seaborn/matplotlib visualization to convert station-month records into an interpretable analytical view.

Interpretation

- State-Level Risk Ranking - What the graph shows: Every state ranked by its average Groundwater Stress Index, from lowest to highest risk.
- High-performing regions: Jammu & Kashmir, Himachal Pradesh, Assam, Meghalaya, and Andhra Pradesh rank lowest-risk - combination of good rainfall, shallow aquifers, and/or lower extraction intensity.
- Low-performing regions: Haryana (0.61), Maharashtra (0.60), Rajasthan (0.58), and Punjab (0.58) top the risk ranking - directly corroborating decades of CGWB "over-exploited block" classifications for this region.
- Geographic clusters: Three of the top four highest-risk states form a contiguous northwest belt, reinforcing the spatial clustering seen in Map 21.

Significance

Spatial analysis converts large monitoring data into actionable hotspots. State, district and basin views help prioritize where water management attention is most needed.

Notebook Visual 22: Graph 21: River Basin-Level Trend

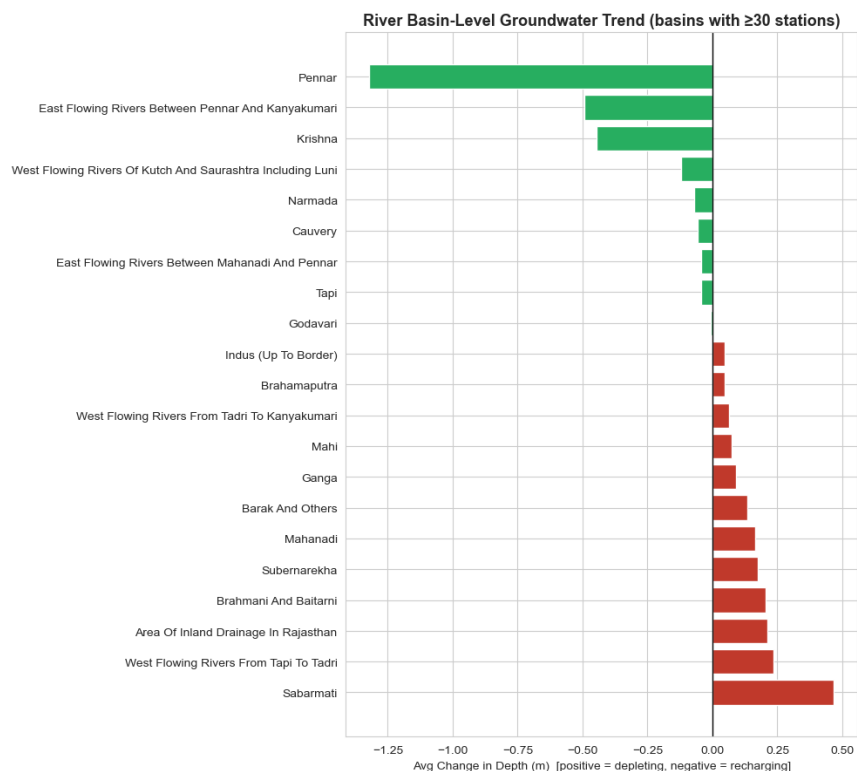


Figure 23: Graph 21: River Basin-Level Trend

Methodology

The notebook groups observations by year, month or season and compares average groundwater level, level change, rainfall and temperature over time.

Interpretation

- River Basin-Level Trend - What the graph shows: Net depletion/recharge trend aggregated by the 17 river basins with sufficient station coverage (≥ 30 stations) for statistical reliability.
- High-performing regions: The Krishna, East Flowing Rivers (Pennar-Kanyakumari), and Pennar basins show the strongest net recharge - likely benefiting from both rainfall and lower extraction intensity.
- Low-performing regions: Sabarmati (Gujarat) shows the steepest depletion trend, followed by West Flowing Rivers (Tapi-Tadri) and the Inland Drainage area of Rajasthan.
- Geographic clusters: Basin-level findings align with the state-level findings - Gujarat and Rajasthan basins reinforce the northwest/west India risk cluster identified at the state level.

Significance

Spatial analysis converts large monitoring data into actionable hotspots. State, district and basin views help prioritize where water management attention is most needed.

8. Trend, Spatial and Risk Analysis

The trend analysis demonstrates that groundwater behavior is strongly seasonal. A negative level_diff indicates a reduction in depth-to-water, which means the water table moved closer to ground level and recharge occurred. A positive level_diff indicates depletion because depth-to-water increased.

Yearly Trend Summary

Year	Avg GW Level (m)	Avg Change (m)	Avg Rain (mm)	Avg Tmax (C)
2019	7.905	-0.263	120.994	31.485
2020	7.029	0.099	95.711	28.387
2021	7.276	0.734	18.904	26.677

Seasonal Summary

Season	Avg GW Level (m)	Avg Change (m)	Avg Rain (mm)	Avg Tmax (C)
Winter (DJF)	7.643	1.04	14.392	26.421
Pre-Monsoon (MAM)	10.327	1.759	27.387	38.326
Monsoon (JJAS)	6.922	-2.641	382.369	31.131
Post-Monsoon (ON)	6.407	-0.432	38.886	29.634

Risk Hotspots

- The dashboard and EDA both identify pre-monsoon as the period with highest average groundwater depth, indicating peak seasonal stress.
- Post-monsoon shows the lowest average groundwater depth, confirming recharge after rainfall.
- Chandigarh appears as the deepest average state in the Power BI outcome dashboard.
- District ranking highlights urban and irrigation-intensive locations where level change is more concerning.

Top Districts by Depletion Rate

State	District	Avg Change	Avg Depth	Stations
Haryana	Gurugram	3.984	34.732	13
Gujarat	Gandhinagar	1.88	91.01	22
Haryana	Rewari	1.869	29.842	10
Punjab	S.A.S Nagar	1.805	10.941	11
Chandigarh	Chandigarh	1.414	25.028	23
Haryana	Nuh	1.244	7.123	14
Gujarat	Banas Kantha	1.145	37.048	41
Rajasthan	Karauli	0.968	15.362	11
Haryana	Jind	0.947	17.75	24
Haryana	Palwal	0.767	9.58	17

9. Power BI Dashboard 1: Climate Drivers of Groundwater

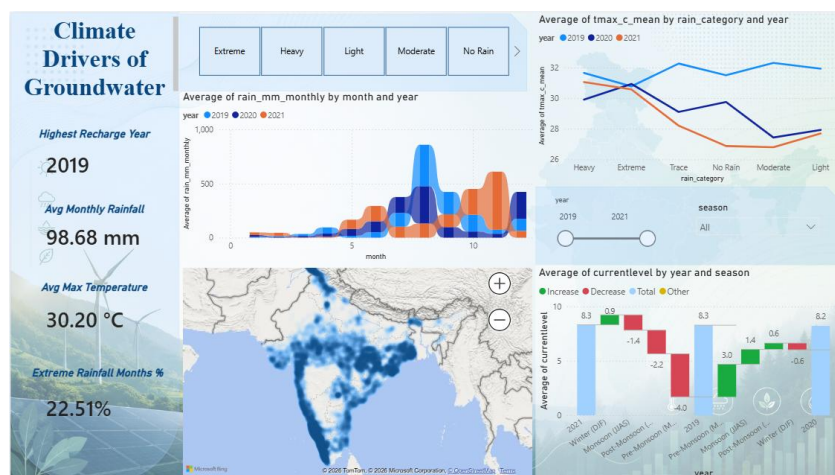


Figure 24: Power BI dashboard page titled Climate Drivers of Groundwater.

Dashboard Purpose

This dashboard explains climate-side drivers that influence groundwater behavior. It combines KPI cards, rain-category slicers, a year range slicer, a season filter, rainfall and temperature charts, a station map and a waterfall-style groundwater-level comparison.

KPI / Visual	Displayed Value	Interpretation
Highest Recharge Year	2019	2019 has the strongest recharge signal based on average level change.
Avg Monthly Rainfall	98.68 mm	Average rainfall from raw Climate_Data records.
Avg Max Temperature	30.20 C	Average of tmax_c_mean in the raw climate table.
Extreme Rainfall Months %	22.51%	Share of climate records labeled Extreme.

Dashboard Map Explanation

What the Dark Blue to Light Blue/White Gradient Means

The map visualizes the spatial distribution of average monthly rainfall across India.

Dark Blue Areas

- Represent locations with higher average rainfall values.
- These regions contribute more strongly to the map density because the average rain_mm_monthly is larger.
- Indicates zones receiving comparatively heavier precipitation.

Medium Blue Areas

- Represent locations with moderate average rainfall.

Light Blue / White Areas

- Represent locations with lower average rainfall values.
- These areas have less influence on the density layer and indicate relatively drier conditions.

Report Description

The Climate Drivers of Groundwater map presents the geographical distribution of rainfall observations across India using latitude and longitude coordinates. The color intensity corresponds to the Average Monthly Rainfall (rain_mm_monthly). Dark blue regions indicate locations experiencing higher rainfall, while lighter blue and white regions indicate lower rainfall levels. This visualization helps identify rainfall concentration patterns that influence groundwater recharge across different regions.

Visual-by-Visual Interpretation

- Rain category buttons at the top act as categorical slicers for Extreme, Heavy, Light, Moderate and No Rain. They allow the user to isolate how different rainfall intensities affect the visuals.
- The monthly rainfall chart compares average rain_mm_monthly across month and year. It shows monsoon concentration and helps identify months of high recharge potential.
- The temperature line chart compares average tmax_c_mean by rain_category and year. It shows that dry or low-rain categories can still carry high temperature stress.
- The India map plots each station using latitude_gd as Latitude and longitude_gd as Longitude. The bubble size is Average of rain_mm_monthly, so larger bubbles represent locations receiving higher average monthly rainfall.
- In the Climate Drivers map, the dark blue gradient indicates higher average rainfall intensity. Areas shown in lighter blue or close to white represent lower average rainfall or weaker rainfall contribution. Therefore, this map depicts spatial rainfall distribution, not groundwater depth.
- The year slider and season dropdown provide temporal filtering. These interactions let users compare climate effects within specific years and seasons.
- The waterfall chart links climate-year-season context to average groundwater level, showing how seasons contribute to increases or decreases in depth-to-water.

Business Insight

The climate dashboard shows that recharge planning must focus on the monsoon window but cannot ignore heat stress. Even where rainfall occurs, high temperature and uneven spatial distribution can reduce effective recharge. The dashboard is useful for climate-risk monitoring, seasonal water planning and communicating where rainfall intensity is not translating into sufficient groundwater improvement.

10. Power BI Dashboard 2: Groundwater Outcomes

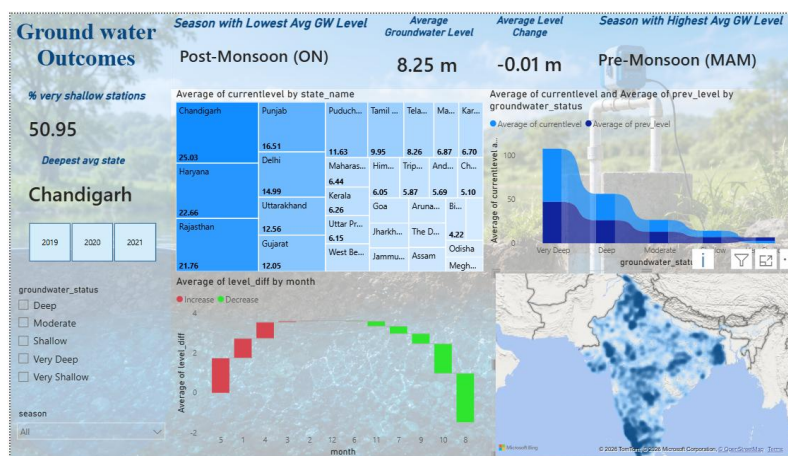


Figure 25: Power BI dashboard page titled Groundwater Outcomes.

Dashboard Purpose

This dashboard focuses on groundwater outcomes rather than climate inputs. It summarizes average groundwater level, level change, status categories, state-level depth, monthly change and geographic distribution.

KPI / Visual	Displayed Value	Interpretation
Season with Lowest Avg GW Level	Post-Monsoon (ON)	Water table is shallowest after monsoon recharge.
Average Groundwater Level	8.25 m	Overall mean currentlevel from groundwater records.
Average Level Change	-0.01 m	Near-neutral average level change across all readings.
Season with Highest Avg GW Level	Pre-Monsoon (MAM)	Water table is deepest before monsoon recharge.
% Very Shallow Stations	50.95	Very Shallow is the largest groundwater status share.
Deepest Avg State	Chandigarh	State/UT with highest average depth-to-water.

Dashboard Map Explanation

What the Dark Blue to Light Blue/White Gradient Means

The map visualizes the spatial distribution of groundwater levels.

Dark Blue Areas

- Represent locations with higher average groundwater levels (currentlevel).
- These areas have greater concentration/intensity on the map.

Medium Blue Areas

- Represent locations with moderate groundwater levels.

Light Blue / White Areas

- Represent locations with lower average groundwater levels.
- These locations contribute less intensity to the density map.

Report Description

The Groundwater Outcomes map displays the spatial variation of groundwater levels across India using station coordinates. The visualization is weighted by the Average Groundwater Level (currentlevel). Dark blue regions indicate locations with relatively higher groundwater levels, whereas lighter blue and white regions indicate lower groundwater levels. The density pattern enables identification of regions with stronger groundwater availability and areas where groundwater levels are comparatively lower.

Visual-by-Visual Interpretation

- The year buttons allow comparison of 2019, 2020 and 2021 groundwater outcomes.
- The groundwater_status checkbox slicer filters Deep, Moderate, Shallow, Very Deep and Very Shallow categories.
- The season slicer filters the entire dashboard by meteorological season.
- The treemap ranks states by average currentlevel. Larger and darker blocks identify states with deeper average water tables.
- The status comparison visual contrasts currentlevel and prev_level by groundwater_status, helping users compare present and previous depth across severity buckets.
- The monthly waterfall chart shows average level_diff by month. Red bars indicate increase in depth-to-water, while green bars indicate decrease and recharge.
- The India map plots each groundwater station using latitude_gd as Latitude and longitude_gd as Longitude. The bubble size is Average of currentlevel, so larger bubbles represent locations where the average depth-to-water is higher.
- In the Groundwater Outcomes map, the dark blue gradient indicates higher Average of currentlevel. Since currentlevel is depth-to-water in meters, darker and larger bubbles represent deeper groundwater levels and therefore more stressed locations. Lighter blue or white areas represent lower Average of currentlevel, meaning the water table is closer to the ground surface and groundwater is comparatively shallower.

Business Insight

The outcome dashboard makes the seasonal groundwater cycle immediately visible: pre-monsoon is the stress period and post-monsoon is the recovery period. This supports decisions such as pre-monsoon demand management, targeted recharge structures, irrigation scheduling and priority monitoring in deeper-state or deeper-district areas.

11. Key Findings and Business Insights

- The relational dataset is clean and analysis-ready, with zero missing cells and zero duplicate rows in all three source sheets.
- The analytical master dataset contains 107,484 station-month records, allowing robust seasonal, temporal and spatial analysis.
- Very Shallow is the dominant groundwater status category in the raw groundwater table, but deep and very deep records identify important stress pockets.
- 2019 is the strongest recharge year by average level change in the station-month analysis.
- Monsoon produces the most important recharge signal, while pre-monsoon represents the highest groundwater stress period.
- Rainfall and groundwater response are related, but correlation is not enough because lag effects, prior depth and regional conditions matter.
- The Power BI dashboards convert technical EDA findings into operational views for climate drivers and groundwater outcomes.

Business and Policy Value

- Water managers can use the dashboards to identify when and where groundwater stress increases.
- Seasonal recharge planning can prioritize pre-monsoon stress and post-monsoon recovery monitoring.
- District and basin ranking can support targeted groundwater management interventions.
- The database design can be extended for future years, additional stations and more climate variables.

12. Recommendations for Reducing Water Stress

- 1. Promote Rainwater Harvesting and Artificial Groundwater Recharge**
Implement rainwater harvesting systems, recharge wells, percolation tanks, and check dams to capture monsoon runoff and increase groundwater recharge, particularly in regions experiencing seasonal water scarcity.
- 2. Regulate Groundwater Extraction in High-Risk Areas**
Introduce stricter monitoring and regulation of groundwater abstraction in districts identified as highly stressed. Sustainable withdrawal limits and periodic assessments can help prevent over-exploitation of aquifers.
- 3. Encourage Water-Efficient Irrigation Practices**
Promote the adoption of drip irrigation, sprinkler irrigation, and precision agriculture techniques to reduce agricultural water consumption, which remains one of the largest contributors to groundwater depletion.
- 4. Strengthen Groundwater Monitoring Systems**
Expand the groundwater monitoring network by integrating IoT-based sensors and automated data collection systems. Real-time monitoring can improve decision-making and provide early warnings for areas experiencing rapid groundwater decline.
- 5. Implement Climate-Responsive Water Resource Planning**
Incorporate rainfall variability, seasonal recharge patterns, and temperature trends into regional water management strategies to improve resilience against climate-induced water stress.
- 6. Prioritize High-Risk Districts for Conservation Measures**
Use district- and basin-level risk rankings to prioritize investments in groundwater recharge structures, conservation programs, and sustainable water management initiatives where they are needed most.
- 7. Increase Public Awareness and Community Participation**
Conduct awareness campaigns encouraging responsible groundwater use, rainwater harvesting, and water conservation practices among households, industries, and agricultural communities to support long-term sustainability.
- 8. Enhance Predictive Analytics for Water Resource Management**
Extend the current PostgreSQL, Python, and Power BI framework by integrating additional datasets such as land use, irrigation demand, crop patterns, and population growth. Applying machine learning models can help predict future groundwater stress and support proactive policy decisions.

13. Conclusion

This project demonstrates a complete end-to-end data analytics and database management workflow for water stress analysis across India. The normalized PostgreSQL schema organizes station, climate and groundwater data cleanly. Python EDA validates data quality, engineers the analytical dataset, explores distributions and relationships, detects outliers, identifies seasonal patterns and builds spatial risk views. Power BI dashboards translate the technical findings into accessible decision visuals.

The most important conclusion is that groundwater stress is seasonal, spatial and multi-factorial. Monsoon rainfall provides recharge, but pre-monsoon depletion, high temperature, regional variation and existing depth-to-water conditions strongly shape outcomes. The project therefore provides both a technical database solution and a practical analytical framework for monitoring groundwater sustainability.

Appendix A. SQL Schema

SQL Database Schema

SQL Query Placeholder

The following SQL was provided in SQL_Querys.sql and used as the PostgreSQL implementation base. The original placeholder requirement [PASTE YOUR SQL QUERY HERE] is fulfilled here with the actual SQL schema and validation queries.

```
CREATE TABLE Location_Master (
    station_id INT PRIMARY KEY,

    state_code VARCHAR(20),
    state_name VARCHAR(100),

    district_code VARCHAR(20),
    district_name VARCHAR(100),

    station_name VARCHAR(255),

    basin VARCHAR(255),
    sub_basin VARCHAR(255),

    latitude_gd DECIMAL(10,6),
    longitude_gd DECIMAL(10,6),

    latitude_imd_rain DECIMAL(10,6),
    longitude_imd_rain DECIMAL(10,6),

    latitude_imd_temp DECIMAL(10,6),
    longitude_imd_temp DECIMAL(10,6),

    rain_lat_diff DECIMAL(10,6),
    rain_lon_diff DECIMAL(10,6),

    temp_lat_diff DECIMAL(10,6),
    temp_lon_diff DECIMAL(10,6),

    imd_matched VARCHAR(20)
);

CREATE TABLE Groundwater_Data (
    gw_record_id BIGINT PRIMARY KEY,

    station_id INT NOT NULL,

    original_id VARCHAR(100),

    record_date DATE,

    year INT,
    month INT,

    season VARCHAR(50),
    source VARCHAR(100),
```

```
currentlevel DECIMAL(12,4),
prev_level DECIMAL(12,4),

level_diff DECIMAL(12,4),

groundwater_status VARCHAR(100),

CONSTRAINT fk_groundwater_station
    FOREIGN KEY (station_id)
    REFERENCES Location_Master(station_id)
);

CREATE TABLE Climate_Data (
    climate_record_id BIGINT PRIMARY KEY,

    station_id INT NOT NULL,

    year INT,
    month INT,

    rain_mm_monthly DECIMAL(12,4),

    rain_category VARCHAR(50),

    tmax_c_mean DECIMAL(8,2),
    tmin_c_mean DECIMAL(8,2),

    temp_range_c DECIMAL(8,2),

    CONSTRAINT fk_climate_station
        FOREIGN KEY (station_id)
        REFERENCES Location_Master(station_id)
);

select * from Location_Master;
select * from Groundwater_Data;
select * from Climate_Data;
select count(*) from Location_Master;
select count(*) from Groundwater_Data;
select count(*) from Climate_Data;
```

Appendix B. Data Dictionary Summary

Location_Master

Column	Detected Type
station_id	int64
state_code	int64
state_name	str
district_code	int64
district_name	str
station_name	str
basin	str
sub_basin	str
Latitude (GD)	float64
Longitude (GD)	float64
Latitude (IMD_rain)	float64
Longitude (IMD_rain)	float64
Latitude (IMD_temp)	float64
Longitude (IMD_temp)	float64
Rain_Lat_Diff	float64
Rain_Lon_Diff	float64
Temp_Lat_Diff	float64
Temp_Lon_Diff	float64
IMD_Matched	bool

Groundwater_Data

Column	Detected Type
gw_record_id	int64
station_id	int64
_id	int64
date	datetime64[us]
year	int64
month	int64
season	str
source	str
currentlevel	float64
prev_level	float64
level_diff	float64
groundwater_status	str

Climate_Data

Column	Detected Type
climate_record_id	int64
station_id	int64
year	int64
month	int64
rain_mm_monthly	float64
rain_category	str
tmax_c_mean	float64
tmin_c_mean	float64
temp_range_c	float64